Using solution x-ray scattering to determine the high-resolution structure and morphology of PEGylated liposomal doxorubicin nanodrugs

Schilt Y.[1,3], Berman T. [2,3], Wei X. [2,3], Barenholz Y.[2,3,*] and Raviv U.[1,3,*]

[1]Institute of Chemistry, The Hebrew University of Jerusalem, Jerusalem, 91904, Israel

[2] Laboratory of Membrane and Liposome Research, IMRIC, The Hebrew University-Hadassah Medical School, Jerusalem, Israel

[3]The Center for Nanoscience and Nanotechnology of The Hebrew University of Jerusalem

* Corresponding authors: uri.raviv@mail.huji.ac.il, chezyb@gmail.com

Supporting Information

1. Background subtraction

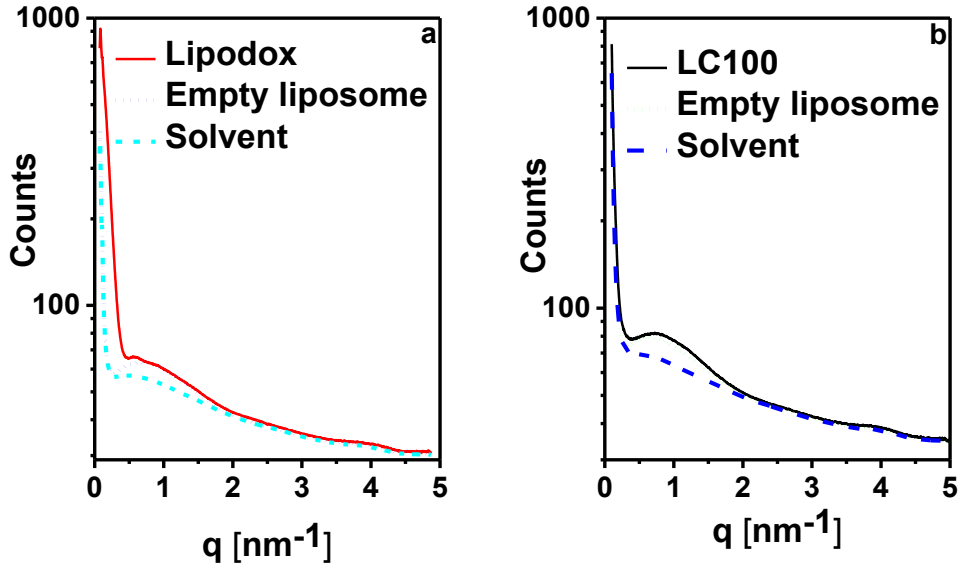


Figure S1a: Radially averaged scattering intensities of Lipodox® (solid red curve), empty liposomes with the same membrane composition and intra-liposomal solution as Lipodox® (dotted blue curve) and a solution of 10 wt% sucrose and 10 mM histidine (the solvent, dashed cyan curve). All measurements were collected under the same conditions at ambient room temperature. The samples were injected into the same flow-through capillary setup, allowing a precise measurement of the capillary and solvent background of each sample. The signal of the solvent was the background signal subtracted from the radially averaged scattering intensities of the empty or loaded liposomes before any further data analysis. The subtracted signal is the scattered intensity presented in Figure 1a. (empty liposome) and Figure 3a. (loaded liposome). **b:** Radially averaged scattering intensities of LC100 (solid black curve), the corresponding empty liposomes with the same membrane composition and intra-liposomal solution (dotted green curve), and a solution of 10 wt% sucrose and 10 mM histidine (the solvent, dashed cyan curve). The measurements were performed using a flow-through capillary setup as in **a**. The subtracted signal is the scattered intensity presented in Figure 9b.

2. **Data reproducibility within the same batch**

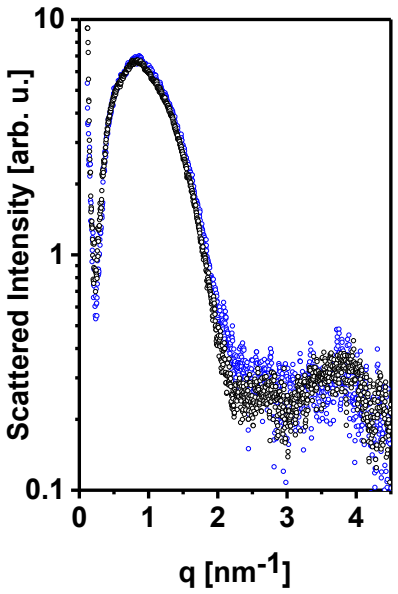


Figure S2: Repeated measurements of empty liposomes used in Doxil®/Caelyx®, Lipodox®, or DOX-NP™ from the same batch taken at 18 months interval. Within the noise, the data are essentially identical except for some small constant background that determines the intensity at high *q* values.

3. Structural fitting parameters of the empty and loaded liposomes

Table S1: Structural parameters obtained from the fit of the scattered intensities (Figure S3) of two separate preparations of the empty liposome used in Doxil®/Caelyx®, identical for Lipodox® and DOX-NP™ to the 5 Gaussian stack model (eq.1). The best fit was found without assuming an "inner medium" layer, suggesting that the electron densities inside and outside the liposome were similar. In other words, the sucrose and histidine solution outside the liposomes had a similar electron density to the salt solution inside the liposomes. Data from preparation 1 are presented in Figure 1a.

	Empty liposomes of Doxil®/Caelyx®, Lipodox®, or DOX-NP™					
	Preparation 1			Preparation 2		
	z_i[nm]	$\Delta\rho_i$[e/nm^3]	τ_i [nm]	z_i[nm]	$\Delta\rho_i$[e/nm^3]	τ_i [nm]
PEG 1	-3.6 ± 0.4	0.8 ± 0.1	3.9 ± 0.6	-3.6 ± 0.4	0.9 ± 0.1	3.5 ± 0.4
Head 1	-2.40 ± 0.03	48.0± 0.1	0.51 ± 0.01	-2.40 ± 0.03	43.4 ± 0.1	0.53 ± 0.01
Tails	0.00	-50.0	1.03 ± 0.01	0.00	-50.0	1.03 ± 0.01
Head 2	2.40 ± 0.03	43.4± 0.1	0.53 ± 0.01	2.40 ± 0.03	47.9± 0.1	0.51 ± 0.02
PEG 2	3.6 ± 0.6	2.8± 0.1	4.6 ± 0.6	3.6 ± 0.6	3.7 ± 0.1	4.9 ± 0.6
L_{in}	3.6 ± 1.2			3.9± 1.2		
L_{out}	4.8± 1.2			5.1± 1.2		

Table S2: Structural parameters obtained by fitting the scattered intensities (Figure 9b,c) of the empty liposomes used in LC100 (prepared in ammonium methanesulfonate) to the calculated scattered intensity from a stack of infinite flat slabs with Gaussian electron density profiles (eq. 1), and the scattered intensity from LC100 to the calculated scattered intensity from the liposome model (eq. 3). Note that the contrast between the inner medium and the inner PEG layer was below our measurement resolution in the case of LC100.

	Empty LC100			LC100		
$\Delta\rho$ Inner medium	0			1.2 [e/nm^3]		
	z_i [nm]	$\Delta\rho_i$ [e/nm^3]	τ_i [nm]	z_i [nm]	$\Delta\rho_i$ [e/nm^3]	τ_i [nm]
PEG 1	-3.6 ± 0.8	1.1 ± 0.1	3.4 ± 0.5	-3.6 ± 0.4	1.2 ± 0.1	5.1± 1.5
Head 1	-2.14 ± 0.03	47.9± 0.1	0.51± 0.01	-2.14 ± 0.03	47.9± 0.1	0.51± 0.01
Tails	0	-50.0	1.03± 0.01	0	-50.0	1.03± 0.01
Head 2	2.55 ± 0.03	43.4± 0.1	0.53± 0.01	2.55 ± 0.03	43.4± 0.1	0.53± 0.01
PEG 2	3.8 ± 0.6	3.3± 0.3	4.3± 0.4	3.2 ± 0.4	4.0± 0.3	4.5± 0.6
L_{in}	4.1 ± 1.3			below the experimental resolution		
L_{out}	4.6 ± 1.0			4.3 ± 1.0		

Table S3: The parameters that best fit the scattered intensity of the empty liposomes which are identical for Doxil®/Caelyx®, Lipodox®, or DOX-NP™, using the liposome model (eq. 3).

	Empty liposomes used in Doxil®/Caelyx®, Lipodox®, or DOX-NP™		
	R_i[nm]	$\Delta\rho_i$[e/nm^3]	τ_i[nm]
PEG 1	36.43	0.81	3.87
Head 1	37.60	47.93	0.51
Tails	40.00	-49.97	1.03
Head 2	42.39	43.38	0.53
PEG 2	43.56	2.76	4.59

4. **Measurement of different batches of empty liposomes used in Doxil®/Caelyx®, Lipodox®, or DOX-NP™**

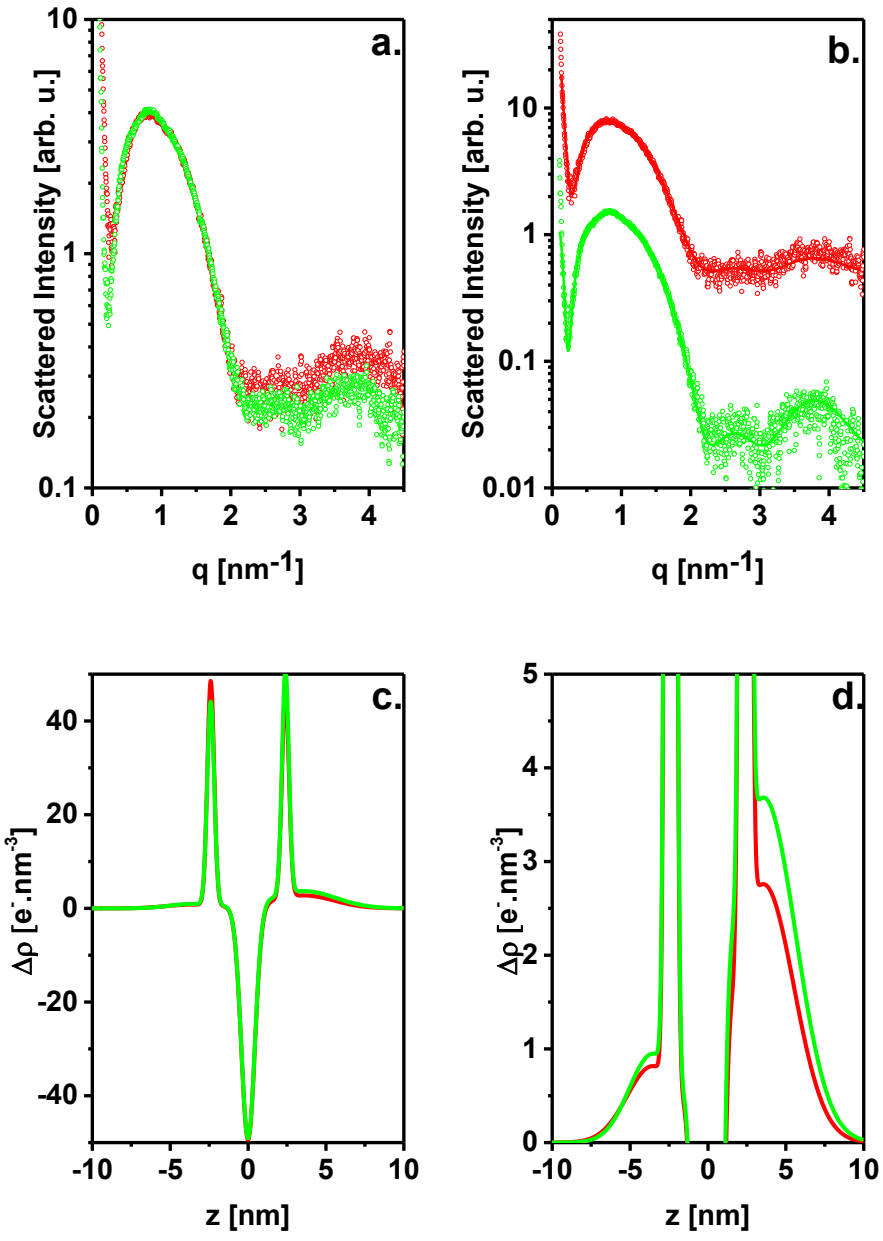


Figure S3 a: Scattered intensity from two batches of empty liposome dispersions used in Doxil®/Caelyx®, Lipodox®, and DOX-NP™ (open symbols). The main difference between the data comes from the noise ratios and the background levels. **b:** The

measured signal was fitted to a model of a stack of five infinite flat slabs having Gaussian electron density profiles along the normal to the membrane plane (continuous line). The data have been shifted for a better understanding. We present in our paper the data with the lower noise level (green symbols). **c:** Electron density contrasts, $\Delta\rho$, of the liposome membrane that best fitted the scattered intensities presented in **a**, as a function of z (the direction normal to the bilayer plane). The electron densities of the three batches are essentially similar, the differences defining the error-bars of the fitting parameters. The parameters that best fit the data are presented in Table S1. **d:** The PEG layer electron density profile is shown on an expanded scale.

5. Effect of the radius polydispersity

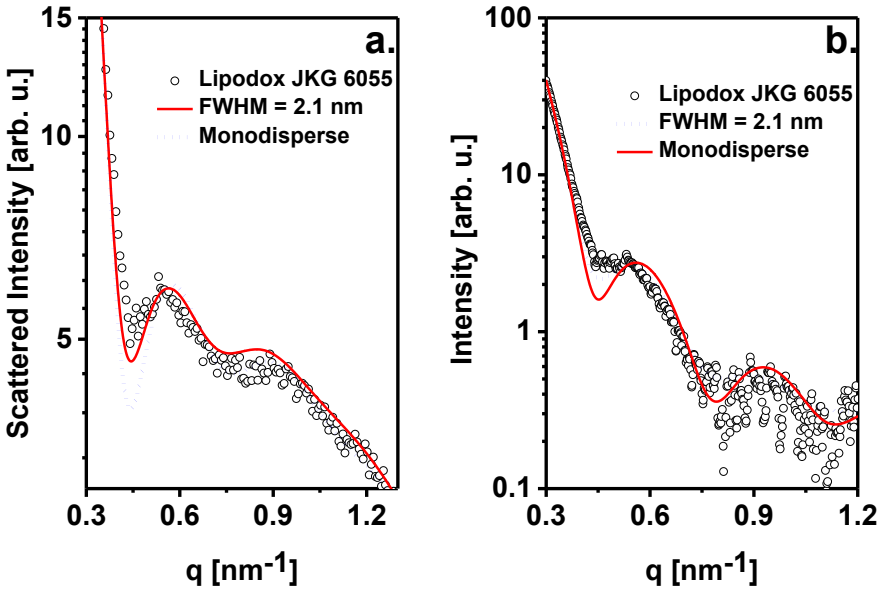


Figure S4 a: Radially integrated scattering intensity from a solution of Lipodox®. The data were fitted to a linear combination of the empty liposome model (Figure 1a) and a cylindrical rod with a length of 70 nm and a radius of 9.1 nm, modeling the doxorubicin crystal. **b:** The difference between the signals Lipodox® #JKG6055 and the model that best fit the data of empty liposomes, shown in Figure 1a. For **a** and **b**, the red solid curve was obtained with a radius polydispersity of FWHM = 2.1 nm, whereas the blue broken curve was obtained by assuming a monodispersed cylindrical rod with the same radius. The polydispersity might have been overestimated as the second oscillation of the model with polydispersity is flattened compared to the data, while the monodispersed model gives a poorer fit for the first minimum, but reproduces well the second oscillation of the data. The overall fit, however, was best with polydispersity of 2.1 nm. Within the framework of our model and without any additional free parameters this seemed to be the best approach.

6. Doxorubicin signal extraction

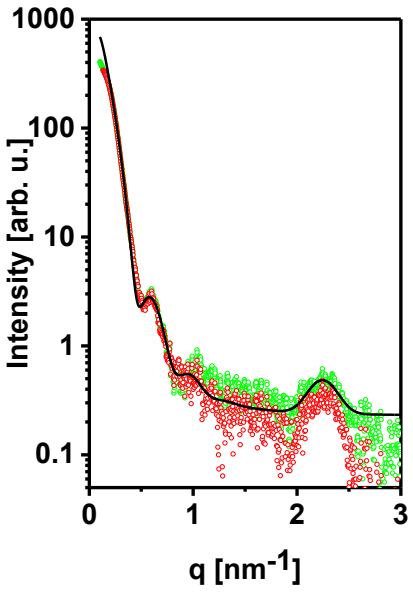


Figure S5: The difference between the signals of Doxil® #1300667 and the models that best fit the data of empty liposomes (green and red open symbols shown in Figure S3a for green and red symbols in this figure). No difference is visible between the two subtractions; hence the details of the PEG layers of the liposomes membranes did not influence the fit of the doxorubicin crystal. The nanocrystal was fitted to a model of a 70-nm-long rod (eq. 2), with a polydispersed radius of average value 8.4 nm and FWHM 2.1 nm.

7. Comparison between empty liposome containing ammonium sulfate solution and ammonium methane sulfonate solution

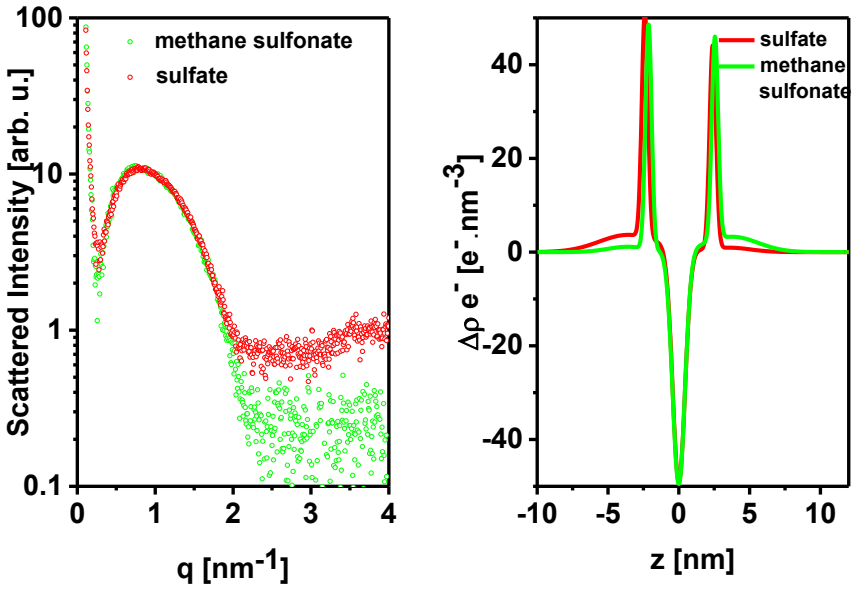


Figure S6 a: Scattered intensities of empty liposome dispersions containing ammonium sulfate (as in Doxil®) and ammonium methane sulfonate (as in LC100). The main difference between the signals is the level of the background. This similarity reinforces the idea that the electron density of the inner medium has no signature in our measurements and that the structure of the membrane was nearly the same in both cases. **b:** Electron densities associated with the data presented in **a**. The head to head distance was almost the same, as well as the thickness of the PEG layers (see Tables S1 and S2).